Rewarding Learning

Centre Number

Candidate Number

General Certificate of Secondary Education
2026

Digital Technology

Unit 4

Digital Development Concepts

[GDG41]

GDG41

TUESDAY 2 JUNE, MORNING

TIME

1 hour 30 minutes.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
You must answer the questions in the spaces provided.
Do not write outside the boxed area on each page or on blank pages.
Complete in black ink only. **Do not write with a gel pen.**
Answer **all twelve** questions.
You **must not** use a calculator for this paper.

INFORMATION FOR CANDIDATES

The total mark for this paper is 120.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in Question **6**.

15023

For Question 1 insert the appropriate letter in the space provided.

1 **(a)** ASCII stands for:

A American Standard Computing for Information Interchange.

B American Standard Code for Information Interchange.

C American Standard Computing for Internet Interchange.

D American Standard Code for Internet Interchange.

Answer _____ [1]

(b) When using 7-bit ASCII for character representation:

A There are a maximum of 128 characters.

B There are a maximum of 127 characters.

C There are a maximum of 256 characters.

D There are a maximum of 255 characters.

Answer _____ [1]

(c) Select the statement that is false about Unicode.

A Unicode can represent more characters than 7-bit ASCII but less characters than 8-bit ASCII.

B Unicode can represent more characters than both 7-bit ASCII and 8-bit ASCII.

C Unicode uses between 8 and 32-bit binary codes to represent characters.

D Unicode contains all of the characters represented in ASCII.

Answer _____ [1]

(d) Select the statement that is true about the file “DigitalTechnology” described below.

Name	Date modified	Type	Size
DigitalTechnology	14/10/2025 21:07	Microsoft PowerPoint Presentation	1024,000 KB

A The file is 10 Megabytes in size.

B The file is 100 Megabytes in size.

C The file is 1000 Megabytes in size.

D The file is 10000 Megabytes in size.

Answer _____ [1]

(e) Complete the table below by inserting the correct unit of data beside each bit pattern. The first one has been completed for you.

Bit Pattern	**Unit of Data**
0	Bit
10101010	[1]
1010	[1]

[Turn over

2 Which letter gives the correct output for each of the following algorithms?

(a)
```
FOR X = 10 TO 15
        OUTPUT X+2
```

A 10 11 12 13 14 15

B 10 12 14 16 18 20

C 12 13 14 15 16 17

D 12 14 16 18 20

Answer _____ [1]

(b)
```
A=0, B=6
WHILE (B>A)
        OUTPUT B
        B=B–2
END WHILE
```

A 6 4 2 0

B 6 4 2

C 2 4 6 8

D 4 2 0

Answer _____ [1]

(c)
```
FOR X = 1 TO 4
        IF (X/4 >= 1)
            FOR Y = 0 TO X
                    OUTPUT "#"
        ENDIF
ENDFOR
```

A # # # # #

B # # # #

C # # #

D #

Answer _____ [1]

[Turn over

3 Jack monitors the temperature of his greenhouse from Monday to Friday. He wants to use a program to process the data collected. He is going to create user-defined functions to solve the problem.

(a) Write **TRUE** or **FALSE** beside each of the following statements about user-defined functions.

Statement	**TRUE/FALSE**
User-defined functions make programs less efficient.	[1]
User-defined functions are re-useable within a program.	[1]
User-defined functions make testing a program easier.	[1]
User-defined functions cannot return values to the main program.	[1]

(b) The program should output a warning message "CODE 1: Too Hot" if the average temperature is more than 25.0°C, otherwise the message should say "CODE 2: No Action Required".

The output message will be stored in a string variable called Message.

(i) Complete the paragraph below about string handling. (Not all words will be used.)

STRING **CONCATENATE** **SUBSTRING** **CHARACTER**

LETTERS **CHARACTERS** **LENGTH** **NUMBERS**

Strings can be joined together using a _____ [1]

function. To extract part of a string a _____ [1]

function can be used. A _____ [1] function can be

used to determine the number of _____ [1]

in a string.

(ii) Complete the algorithm which will find the average of the 5 temperatures t1, t2, t3, t4, and t5 and output the correct message.

average = (t1 + t2 + t3 + t4 + t5)/5.0

OUTPUT average

IF (_____) [3]

Message = "CODE 1: Too Hot"

ELSE

_____ [2]

ENDIF

OUTPUT _____ [1]

(c) Jack has used a Software Development Environment (SDE) to create the program. The program code contains syntax errors.

(i) Explain the term syntax error when referring to program code.

_____

_____ [2]

(ii) Explain how the following two features of a SDE can help Jack when writing code.

Auto Indentation _____

_____

_____

Code Completion _____

_____

_____ [4]

[Turn over

4 **(a)** Convert the denary number 50 to an 8-bit binary pattern.

Show all working out clearly and write your answer in the space provided below.

Answer _____ [2]

(b) Convert the binary pattern 00101111 to:

(i) a hexadecimal value

Show all working out clearly and write your answer in the space provided below.

Answer _____ [2]

(ii) a denary value

Show all working out clearly and write your answer in the space provided below.

Answer _____ [2]

(c) Complete the calculation by entering the correct values in the shaded boxes. Place a circle around any overflow generated.

		1	0	0	1	0	1	0	0
+		1	1	0	0	0	1	1	1
Carry		–	–	–	–		–	–	–
Answer		0	1	0	1	1		1	1

[4]

[Turn over

5 **(a)** Complete the table below by matching the description to the appropriate Boolean operator.

OR **AND** **NOT**

Description	**Boolean operator**
The output is true only if all inputs are true.	[1]
The output is true if one or all of the inputs are true.	[1]
The output is true if the input is false and the output is false if the input is true.	[1]

(b) In the truth table below the output C is 0 only if both inputs A and B are equal to 1.

Insert the Boolean expression for C and complete the truth table.

A	B	C= [2]
0	0	[1]
0	1	[1]
1	0	[1]
1	1	0

6 An array structure can be used to store data.

By referring to the array ***marks*** below, which contains a set of GCSE coursework marks, explain:

- how an array is created or declared;
- how individual elements in the array can be accessed and changed.

marks

60	59	50	55	56	58	60	60	52	50

Quality of written communication will be assessed in this question.

[6]

[Turn over

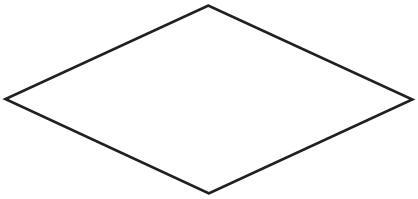
7 **(a)** Name each of the flowchart symbols shown below.



1. _____ [1]

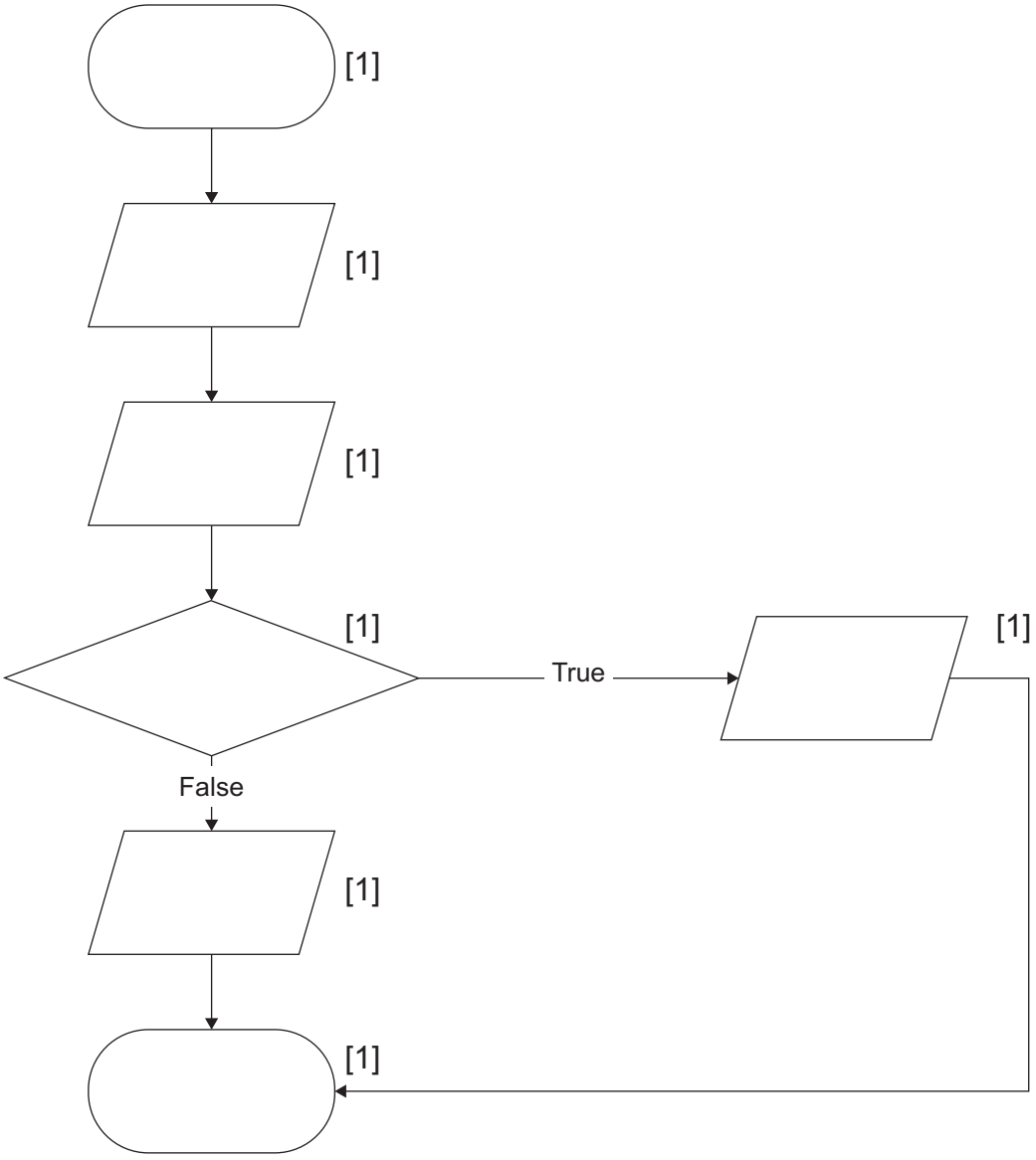


2. _____ [1]



3. _____ [1]

(b) Complete the flowchart below which allows a user to input two numbers, n1 and n2 and outputs the smallest of the two numbers.



[Turn over

(c) Name and draw the flowchart symbol which is used to represent calculations on a flowchart.

Name of symbol	Drawing of symbol
[1]	[1]

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(Questions continue overleaf)

[Turn over

8 A program is required to allow Henry, the manager in a gym, to log into an application.

(a) The programmer is going to use abstraction and decomposition when designing the solution. Explain the term abstraction.

_____

_____ [2]

(b) Pseudo-code will be used to represent the solution. In the table below, write **TRUE** or **FALSE** beside each of the statements about pseudo-code.

Statement	**TRUE/FALSE**
Pseudo-code is a graphical representation of a solution.	[1]
Pseudo-code uses keywords and control structures in a similar way to a programming language.	[1]
Pseudo-code is language independent.	[1]
Pseudo-code is the source code for a solution.	[1]

(c) Henry has been given the username HENRYK156 and must set a password. The criteria for a valid password is listed below.

- The password must begin with 3 letters followed by 3 numbers
- The password must be at least 8 characters long
- The password must contain at least 1 special symbol

He has chosen the password ADM967r!

(i) Complete the table below, by inserting the name of suitable validation checks that could be used to ensure the password meets the criteria.

Criteria	**Validation check**
The password must begin with 3 letters followed by 3 numbers.	[1]
The password must be at least 8 characters long.	[1]

(ii) Complete the part of the algorithm that will check Henry's username and password. The algorithm should output a message to tell him that he has logged in successfully or that his login details are not correct.

```
OUTPUT "Enter username"
INPUT username
OUTPUT "Enter password"
INPUT password
```

[5]

[Turn over

(d) All usernames and passwords are stored in a file.

(i) Why is it necessary to store these details in a file?

_____

_____ [2]

(ii) Apart from opening and closing the file, list two other file handling functions that should be used in a program when adding a new username and password to the file.

1. _____

2. _____ [2]

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(Questions continue overleaf)

[Turn over

9 Martin runs a small dog grooming business.

Grooming costs are listed below:

Dog Size	**Grooming Cost(£)**
Large (L)	80.00
Medium (M)	65.00
Small (S)	45.00

- Dogs can be collected from their owner's home, provided it is not more than 5 miles away. Dog collection costs £6.00.

A program is required which will calculate and output the total cost of the grooming and collection services. The program should output an error message if a dog is to be collected and the owner lives more than 5 miles away. The variables to be used in the program are listed below:

- **DogSize** will store the size of the dog as 'S' for small, 'M' for medium and 'L' for large.
- **DogCollected** will store a value "YES" or "NO" indicating whether or not a dog is to be collected from home.
- **DogDistance** will store the distance between the owner's home and the groomer.
- **GroomCost** will store the cost of grooming the dog.
- **TotalGroomCost** will store the total amount of money that an owner has to pay.

(a) Complete the table below by suggesting the most appropriate data type for the variable listed.

Variable	**Data type**
DogSize	[1]
GroomCost	[1]

(b) Complete the part of the algorithm which will set the variable GroomCost to the correct value based on the dog's size.

```
OUTPUT "Enter the size of the dog S, M, L"

INPUT DogSize
```

[3]

[Turn over

(c) Assuming the variable GroomCost has been set to the correct value, complete the part of the algorithm which will output the correct total cost for the services provided.

```
TotalGroomCost=GroomCost

OUTPUT "Is dog collection needed?"

INPUT DogCollected

IF DogCollected = _____ [1]

    OUTPUT _____ [1]

    INPUT _____ [1]

    IF _____ [3]

        TotalGroomCost= _____ [1] + 6.00

    ELSE

        OUTPUT _____ [1]

    ENDIF

ENDIF

OUTPUT TotalGroomCost
```

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(Questions continue overleaf)

[Turn over

10 Sorting algorithms are used to sort data into a particular order. An array called ***numbers*** stores the following values:

numbers

20	5	40	10	30

(a) Using the Insertion Sort, demonstrate how the data in ***numbers*** would be organised from smallest to largest during the sorting process. Show the array content after each number has been compared to those already sorted.

20	5	40	10	30

[4]

(b) The bubble sort can also be used to sort the data in the array ***numbers***.
Complete the algorithm which represents an efficient solution to the bubble sort.
The size of the array is stored in a variable called **n** and is set to 5.

```
n=5
FOR X = 0 TO _____                                    [1]

    FOR Y = 0 TO _____                                [2]

        IF numbers[Y] > numbers[ _____ ]              [1]

            Temp = numbers[Y]

            numbers[Y] = numbers[Y+1]

            numbers[Y+1] = _____                      [1]

        END IF

    END FOR

END FOR
```

[Turn over

11 A programmer has written a section of code which is not working as expected. The program should check to ensure that a user has entered a value in the range 17 to 20.

(a) She is going to use white box testing to identify the problem with the code. What is white box testing?

_____

_____ [2]

(b) A section of the programmer's test plan for the variable value is shown below.

Complete the test plan.

Test Number	Reason for test	Test data	Expected outcome
1.	[1]	18	Value Accepted
2.	[1]	Press Enter Key	Value Rejected
3.	Extreme Value	[1]	Value Accepted
4.	[1]	35	[1]

(c) Below is an algorithm representing the part of the program that is not working.

1.	valid = FALSE
2.	WHILE valid = FALSE
3.	OUTPUT "Enter a value"
4.	INPUT value
5.	IF (value>=17 OR value <20)
6.	valid = TRUE
7.	ELSE
8.	OUTPUT "Enter a value between 17 and 20"
9.	ENDIF
10.	END WHILE

(i) Complete the sentence below by placing the number of the incorrect line in the space provided.

Line number _____ is incorrect. [1]

(ii) Write the correct version of the algorithm for the incorrect line.

_____ [2]

[Turn over

12 Evaluation of a computer system should be carefully planned.

(a) List two groups of people who should participate in the evaluation of a system.

1. _____

2. _____ [2]

(b) State how the content of each of the following can support evaluation.

(i) documented results from testing

_____

_____ [2]

(ii) user requirements

_____

_____ [2]

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For Examiner's use only	
Question Number	**Marks**
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Total Marks	

Examiner Number

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